

Olympiad Test Paper — Social Science (Class 7) — Paper 3

Chapter 3: Climates of India

Paper 3 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	3 — Climates of India
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Reward identifying the controlling factor (latitude, altitude, continentality, relief, winds) behind each climate fact rather than the surface observation, and refuse the lures of confusing weather with climate, blaming nearness to the Sun for equatorial heat, and ignoring altitude or the moderating sea. Do not write on this paper — mark answers on your answer sheet.

1. A newspaper headline reads, "Heavy showers lash Pune this afternoon." A textbook states, "Pune enjoys a moderate, fairly dry climate." Which word in these two lines is being used correctly, and why?

- (A) Only the headline is correct, because climate is just today's sky and a textbook cannot know it.
- (B) Only the textbook is correct, because a single afternoon of rain is itself the climate of Pune.
- (C) Both are correct: the headline reports one afternoon's condition (weather), the textbook reports the long-term average (climate).
- (D) Both are wrong, because rain can only be described as climate and dryness only as weather.

2. Two friends argue. Riya says her city's climate changed because it rained today after a dry week. Decide who is reasoning correctly.

- (A) Riya is right, because every day of rain shifts the climate by a measurable amount.
- (B) Riya is right, because climate and weather mean exactly the same thing.

(C) Riya is wrong: a single day's rain is weather, and climate is judged only over many years, so one rainy day cannot change a climate.

(D) Riya is wrong only because it did not rain enough; heavier rain would have changed the climate.

3. Singapore (near the Equator) and Delhi (well to the north) are compared. Singapore is warm and even all year while Delhi has scorching summers and chilly winters. The single factor most responsible for Delhi's wider swing is:

(A) its lower altitude compared with Singapore

(B) Singapore being far closer to the Sun than Delhi

(C) Delhi having no atmosphere to trap heat

(D) its higher latitude, which gives a stronger seasonal change in how directly the Sun's rays fall

4. A student claims, "Places on the Equator are hot simply because they are nearer the Sun." The best correction is that equatorial places are hot because:

(A) the Equator is several million kilometres closer to the Sun than the poles

(B) sunlight at the Equator never has to pass through any air

(C) the Earth's core heats the Equator from below

(D) the Sun's rays there fall almost vertically, so the same energy is concentrated on a small area

5. Town X sits at 1,800 m on a plateau; Town Y lies at 200 m in a nearby valley at the same latitude. Reading their average temperatures, you would expect:

(A) Town X to be warmer, because higher places are nearer the Sun

(B) both to be identical, since they share the same latitude

(C) Town X to be distinctly cooler, because temperature falls as altitude rises

(D) Town Y to be cooler, because valleys are always closer to the poles

6. Why do the high Himalayan peaks stay snow-clad through the year while the plains a short distance away swelter, even though both lie at similar latitudes?

(A) The peaks are at a much higher latitude than the plains.

(B) The plains are far closer to the sea, which heats them.

(C) Winds from the Equator warm only the plains and never the peaks.

(D) Great altitude on the peaks keeps the air thin and cold, so temperature there stays far below the warm plains.

7. Chennai and Hyderabad lie at nearly the same latitude, but Chennai's yearly temperature range is much smaller than Hyderabad's. The most direct reason is that Chennai:

(A) is much higher in altitude than Hyderabad

(B) lies far nearer the Equator than Hyderabad

(C) receives no sunshine in summer

(D) is on the coast, where the slow-heating, slow-cooling sea evens out the temperature

8. An interior city on a high plateau records a very large gap between its hottest and coldest readings. Which combination of factors best explains such a wide temperature range?

- (A) Being far from the moderating sea (continentality) combined with greater height both work to widen the swing.
- (B) Nearness to the sea plus low altitude.
- (C) Lying exactly on the Equator with constant overhead Sun.
- (D) Frequent moist sea winds blowing over it all year.

9. The whole Indian monsoon system ultimately rests on one physical fact about land and water. That fact is:

- (A) the Moon's gravity drags monsoon clouds toward India each summer
- (B) rivers push warm air upward and pull in sea winds
- (C) the Equator stays at one fixed temperature all year
- (D) land heats up and cools down faster than the ocean, so pressure differences build up between them

10. In summer the interior of Asia becomes intensely hot. The pressure that develops there, and the wind it produces, are:

- (A) high pressure, which pushes dry winds out to sea
- (B) low pressure, into which dry winds blow out toward the sea
- (C) low pressure, into which moisture-laden sea winds blow from the south-west
- (D) high pressure, which draws moist sea air inland

11. The summer monsoon is called the "south-west" monsoon. The compass term refers to:

- (A) the direction from which the rain-bearing winds arrive, blowing in from the south-west
- (B) the part of India that the rains avoid
- (C) the direction in which the rivers flow during the rains
- (D) the corner of the sky where the Sun sets in summer

12. Why does the south-west monsoon bring India's main rains while the winter monsoon does not?

- (A) Summer winds come from the dry Thar while winter winds come from the sea.
- (B) Summer winds are colder, and cold winds always carry more rain.
- (C) Summer winds blow straight down from the upper atmosphere carrying rain.
- (D) Summer winds cross warm seas and arrive moisture-laden, while winter winds blow off the dry land toward the sea.

13. The sudden, dramatic onset of heavy monsoon rain after the dry pre-monsoon heat is popularly called the "burst" of the monsoon. The best description of why it feels so sudden is that:

- (A) moist sea winds arrive in force and unleash steady heavy rain after weeks of building heat and dryness
- (B) the Sun suddenly stops shining over India
- (C) the rivers suddenly rise and spill into the air as rain
- (D) the winter monsoon arrives all at once from the north-east

14. The monsoon usually reaches Kerala in early June and then spreads. In which order does it typically advance across the country?

- (A) from the southern tip northward, covering the whole country by about mid-July
- (B) from the far north southward, reaching Kerala last in September
- (C) from the east coast outward to the seas
- (D) everywhere at exactly the same moment in early June

15. If, in some imaginary year, land and ocean warmed at exactly the same rate, what would happen to the monsoon?

- (A) It would grow twice as strong from the extra even heating.
- (B) It would not form, because the land-sea pressure difference that drives it would vanish.
- (C) It would still arrive but only in winter.
- (D) It would reverse direction and blow from land to sea all year.

16. A meteorologist notes that monsoon winds always move toward the hot Asian interior in summer. The underlying rule she is applying is that air moves:

- (A) from low pressure to high pressure
- (B) from regions of high pressure to regions of low pressure
- (C) always from north to south
- (D) always toward wherever the Sun is overhead

17. In winter the north-eastern monsoon blows out from the land toward the sea. The pressure pattern that produces this is:

- (A) low pressure over the cold land, drawing sea air inland
- (B) equal pressure everywhere, so winds blow at random
- (C) high pressure over the sea, pushing moist air onto the land
- (D) high pressure over the cold land and lower pressure over the warmer sea, so air flows seaward

18. Most of the north-east winter monsoon is dry, yet Tamil Nadu's south-east coast gets notable rain at that time. The reason is that the winter winds reaching it have:

- (A) crossed the Thar Desert and gathered moisture there
- (B) crossed the Bay of Bengal and picked up moisture before reaching the coast
- (C) blown straight down the Himalayas carrying snow-melt
- (D) come fresh off the Arabian Sea from the west

19. The monsoon winds reverse between summer and winter. The deepest reason for this reversal is that:

- (A) the Earth changes the tilt of its axis twice each year
- (B) the Western Ghats physically swing around with the seasons
- (C) the land-sea temperature and pressure pattern flips between the two seasons
- (D) the Moon switches the direction of its orbit

20. The "retreating monsoon" season refers to the period when:

- (A) the south-west monsoon first bursts over Kerala
- (B) the south-west monsoon withdraws and the weather transitions toward the dry winter pattern
- (C) the rivers retreat back into the Himalayas
- (D) the summer heat first builds up before any rain

21. Place these monsoon phases in their correct order through the year: dry winter, hot dry summer, rainy south-west monsoon, retreating monsoon. The correct sequence is:

- (A) dry winter → hot dry summer → rainy south-west monsoon → retreating monsoon
- (B) rainy south-west monsoon → dry winter → hot dry summer → retreating monsoon
- (C) retreating monsoon → rainy south-west monsoon → dry winter → hot dry summer
- (D) hot dry summer → dry winter → retreating monsoon → rainy south-west monsoon

22. India is often said to have a "fifth season" beyond the usual four. Why does India's classification add the rainy season so prominently?

- (A) Because the monsoon rains dominate the year so strongly that they form a distinct season of their own.
- (B) Because India copied a five-season calendar from Europe.
- (C) Because India has no winter at all, leaving room for a fifth.
- (D) Because every Indian festival needs its own separate season.

23. The six traditional Indian *ritus* divide the year more finely than the four-season scheme chiefly because they:

- (A) were invented only recently for tourism
- (B) ignore the monsoon and follow the Sun alone
- (C) match exactly the four European seasons renamed
- (D) track the monsoon-ruled year more closely, adding seasons the four-fold scheme misses

24. Why do the same seasons return year after year in India?

- (A) Because the Sun grows hotter every summer and cooler every winter.
- (B) Because the monsoon winds decide to reverse on fixed festival dates.
- (C) Because rivers flood at the same time each year and create the seasons.
- (D) Because the Earth revolves around the Sun on its tilted axis, repeating the pattern annually.

25. The hot, dry summer wind that sweeps Rajasthan and Punjab, known as the "loo," is so scorching because it has:

- (A) crossed the cool Arabian Sea and absorbed its moisture
- (B) blown down off the snow-covered Himalayas
- (C) crossed hot, dry desert lands and taken on their heat and dryness
- (D) risen straight up from underground hot springs

26. A cold wave grips the north Indian plains in winter when chilly winds spill over the Himalayas. This illustrates the general principle that winds influence a region's climate by:

- (A) physically changing the latitude of the region they reach
- (B) lifting the region to a higher altitude
- (C) carrying the temperature and moisture qualities of the lands they have travelled over
- (D) moving the region nearer to the sea

27. A shaded forest clearing stays cooler and damper than the open, sun-baked field beside it. This small pocket with its own distinct conditions is best called a:

- (A) rain shadow
- (B) microclimate
- (C) ritu
- (D) monsoon

28. A dense concrete city centre stays several degrees warmer at night than the green fields around it. The best label for this difference is a:

- (A) human-made microclimate (urban heat island)
- (B) rain shadow created by tall buildings
- (C) new ritu of the city
- (D) cyclone forming over the streets

29. The Thar Desert suffers searing, dry days while the lands sheltered behind the Himalayas are spared the worst of the icy Central-Asian winds. Taken together, these two facts show the climate-shaping role of:

- (A) latitude alone
- (B) distance from the sea alone
- (C) relief (topography), since both flat openness and mountain barriers shape the local climate
- (D) the position of the Sun alone

30. Why is the region beyond the high Himalayas, such as Ladakh, described as a "cold desert"?

- (A) It is hot and dry like the Thar, just renamed.
- (B) It is dry because the mountains block the rain-bearing winds, and cold because of its great altitude.
- (C) It is cold only because it lies very near the North Pole.
- (D) It is dry because the sea is too close and pulls away all the moisture.

31. Cherrapunji and Mawsynram in the north-east receive among the heaviest rainfall on Earth. The chief reason is that the moist monsoon winds there are:

- (A) blowing off the dry Thar and somehow carrying that desert's rain
- (B) descending the leeward slopes and warming as they fall
- (C) cold winter winds off the Himalayas dropping snow as rain
- (D) funnelled and forced to rise against the hills, cooling and dropping enormous rain

32. The west coast of India is drenched by the summer monsoon while the Deccan Plateau just inland stays comparatively dry. The factor responsible is:

- (A) the Western Ghats, which force the moist winds up and rain on their windward (western) slopes, leaving the leeward Deccan dry
- (B) the Deccan lying much nearer the Equator
- (C) the west coast standing at a far higher altitude
- (D) the Deccan having no rivers to make rain

33. Orographic rainfall on a mountain's windward side occurs because, as moist air is forced to rise, it:

- (A) expands and cools, can hold less water vapour, so the vapour condenses and falls as rain
- (B) is compressed and warms, holding more vapour with no rain
- (C) speeds up and instantly dries without cooling
- (D) sinks back down and rains on the same side it rose

34. The leeward (sheltered) side of a mountain range is typically dry. This dry zone is called a rain shadow because:

- (A) the mountain's shadow physically blocks the sunlight needed for rain
- (B) rain refuses to fall where it is too shady
- (C) the air has already lost its moisture as rain on the windward side and arrives dry on the lee side
- (D) the leeward side faces directly into the moist wind

35. If the Western Ghats were somehow removed entirely, the most likely change to India's west coast would be that it:

- (A) received much more rain than before
- (B) received much less rain, since there would be nothing to lift and wring out the moist winds
- (C) turned instantly into a frozen tundra
- (D) experienced no change of any kind

36. Kanniyakumari (far south, by the sea) and Srinagar (far north, in the mountains) differ sharply in climate. Which two factors most powerfully combine to make Srinagar so much colder?

- (A) its higher latitude and its great altitude both push temperatures down
- (B) its lower latitude and nearness to the sea

- (C) its position right on the Equator and constant overhead Sun
- (D) the moist sea winds that blow over it all year

37. Assertion: Mumbai has milder summers and winters than Nagpur. Reason: the sea heats and cools more slowly than land, moderating nearby places. Choose the best evaluation.

- (A) Both are true, but the reason does not explain the assertion.
- (B) Both the assertion and the reason are true, and the reason correctly explains the assertion.
- (C) The assertion is true but the reason is false.
- (D) The assertion is false but the reason is true.

38. Assertion: Places near the Equator have only small temperature changes through the year. Reason: near the Equator the Sun stays nearly overhead all year. Choose the best evaluation.

- (A) Both are true, but the reason does not explain the assertion.
- (B) The assertion is true but the reason is false.
- (C) Both the assertion and the reason are false.
- (D) Both the assertion and the reason are true, and the reason correctly explains the assertion.

39. Assertion: The winter monsoon brings little rain to most of India. Reason: in winter the winds blow from the sea onto the land. Choose the best evaluation.

- (A) Both the assertion and the reason are true and the reason explains it.
- (B) The assertion is true but the reason is false; in winter the winds actually blow from the land out to the sea.
- (C) Both the assertion and the reason are false.
- (D) The assertion is false but the reason is true.

40. Two coastal cities sit at the same latitude, but one has a slightly larger temperature range than the other. A reasonable extra factor to investigate would be:

- (A) differences in altitude or in how exposed each is to moderating sea winds
- (B) which city is closer to the Sun
- (C) which city the Moon passes over more often
- (D) which city has more festivals

41. A class plots average July rainfall across India and finds the west coast soaked, the Deccan moderate, and the north-east extremely wet. The single idea that best ties all three observations together is:

- (A) latitude alone, since rainfall simply rises toward the Equator
- (B) altitude alone, since higher always means wetter

- (C) relief lifting moist monsoon winds: heavy rain where winds are forced up, less in the rain shadow behind
- (D) distance from the sea alone, since inland is always wettest

42. "Monsoon failure" means the seasonal rains are weak or insufficient. The most likely chain of consequences for India is:

- (A) rivers overflow → widespread floods → bumper harvests
- (B) crops wither → water grows scarce → harvests shrink → food prices rise
- (C) winters turn much warmer → glaciers grow → more water
- (D) the six ģitus vanish → the calendar collapses

43. Trace the chain from a distant cause to a local effect: "warm Pacific water shifts → Indian monsoon weakens → ____." The most likely next link is:

- (A) India receives record floods that year
- (B) the Himalayas grow several metres taller
- (C) rainfall over India falls short, stressing crops and water supplies
- (D) the sea stops moderating the coastline

44. Why does a weak monsoon hit India's economy so hard?

- (A) Because festivals are cancelled by government order during dry years.
- (B) Because the sea stops moderating the coastal cities.
- (C) Because a large share of farming depends on monsoon rain, so poor rains cut harvests and raise food prices.
- (D) Because the Himalayas block trade when it does not rain.

45. Every few years the monsoon is disrupted by a warming of the central and eastern Pacific Ocean that rearranges tropical winds. This recurring event is known as:

- (A) El Niño
- (B) the loo
- (C) a rain shadow
- (D) the burst

46. Which everyday example best shows the same physics that drives the summer monsoon?

- (A) A ball rolling downhill under gravity.
- (B) A daytime sea breeze, where warm land draws in cooler air off the sea.
- (C) The Moon raising ocean tides.
- (D) A river flowing from the mountains to the sea.

47. The festivals Bihu, Pongal, Onam and Baisakhi are tied to the seasons and the farming cycle. What does this connection reveal about climate?

- (A) Festivals control the climate by deciding when the rains arrive.
- (B) Climate, through the monsoon and seasons, shapes farming and therefore the timing of cultural life.
- (C) Climate has no effect on human culture in India.
- (D) Festivals depend only on each state's latitude.

48. The word "monsoon" comes from the Arabic mausim, meaning "season." Strictly, the term names:

- (A) the winds that reverse direction with the seasons, not the rain by itself
- (B) the rainfall alone, with the winds being unrelated
- (C) the Indian Ocean itself
- (D) the set of six ritus

49. A cyclone forms over warm seas when the central air pressure becomes:

- (A) much higher than its surroundings, pushing air away
- (B) much lower than its surroundings, so air and moisture rush in and spiral upward
- (C) exactly equal to its surroundings everywhere
- (D) colder than the land in the middle of winter

50. During a cyclone the wind suddenly drops and the sky clears overhead. The safest interpretation is that:

- (A) the storm has fully ended and it is safe to go outside
- (B) the cyclone has changed into a harmless breeze
- (C) the eye is the most destructive part and one must flee through it
- (D) the calm eye is passing over, and fierce winds will return from the opposite direction

51. Where in a cyclone are the most violent winds found?

- (A) in the calm, clear eye at the very centre
- (B) hundreds of kilometres out beyond the storm
- (C) in the dense ring of cloud (the eyewall) surrounding the calm centre
- (D) only after the storm has completely passed

52. Burning fossil fuels is said to thicken Earth's natural "greenhouse blanket." The correct cause-and-effect is that the extra greenhouse gases:

- (A) trap more outgoing heat, so the planet warms
- (B) let more heat escape, so the planet cools
- (C) block incoming sunlight, so the planet freezes
- (D) remove the atmosphere, so there is no climate at all

53. Place these links in order for the chain "a factory burns coal → a farmer's crop fails":

- (A) coal burnt → sea cools → monsoon strengthens → crop thrives
- (B) coal burnt → carbon dioxide released → extra heat trapped → climate shifts → monsoon disrupted → crop fails
- (C) coal burnt → rivers freeze → no irrigation → crop fails
- (D) coal burnt → forests grow → more rain → crop floods

54. Two responses to climate change are "mitigation" and "resilience." Which pairing is correct?

- (A) Mitigation builds flood walls; resilience cuts emissions.
- (B) Both simply mean planting more trees and nothing else.
- (C) Mitigation ignores the cause while resilience ignores the effects.
- (D) Mitigation slows the warming (for example by cutting emissions); resilience helps communities withstand and recover from impacts.

55. Which set of actions counts as climate "mitigation"?

- (A) Clearing more forests for farmland.
- (B) Expanding renewable energy and planting trees to cut and absorb greenhouse gases.
- (C) Burning extra coal for cheaper electricity.
- (D) Encouraging wasteful, high-consumption lifestyles.

56. Since the nineteenth century, the main driver of accelerating climate change has been:

- (A) natural processes acting entirely on their own
- (B) the Earth slowly drifting farther from the Sun
- (C) volcanic eruptions and nothing else
- (D) human activities, especially the burning of fossil fuels

57. Why does the natural greenhouse effect actually make life on Earth possible?

- (A) Because they block all of the Sun's rays, keeping Earth cool.
- (B) Because they cool the planet to below freezing.
- (C) Because greenhouse gases trap some of the Sun's heat, keeping Earth warm enough for living things.
- (D) Because they create the monsoon winds directly.

58. A traveller crosses from the rainy windward slope of a hill to the dry valley on the far side within a few kilometres. The rapid change is best explained by:

- (A) a sudden jump in latitude over those few kilometres
- (B) the valley being far closer to the Equator
- (C) the windward slope being made of different soil
- (D) the rain shadow effect: moist air rains on the windward slope and arrives dry on the leeward side

59. India experiences many different climates at the same moment. The most fundamental reason is that India is:

- (A) large and varied, spanning a wide range of latitude, altitude and distance from the sea
- (B) ruled by many different calendars in different states
- (C) lit by a Sun that shines differently on each state
- (D) made of states that each control their own weather

60. Match each region to its dominant climate factor: (i) Ladakh cold desert, (ii) Mumbai's mild range, (iii) Cherrapunji's heavy rain, (iv) Kanniyakumari's year-round warmth. The best matching is:

- (A) (i) low latitude, (ii) high altitude, (iii) distance from the sea, (iv) relief
- (B) (i) nearness to the sea, (ii) latitude, (iii) altitude alone, (iv) winds only
- (C) (i) altitude + rain shadow, (ii) nearness to the sea, (iii) relief lifting moist winds, (iv) low latitude
- (D) (i) winds only, (ii) rain shadow, (iii) low latitude, (iv) altitude

Solutions — Social Science (Class 7), Chapter 3: Climates of India — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	A	31	D	41	C	51	C
2	C	12	D	22	A	32	A	42	B	52	A
3	D	13	A	23	D	33	A	43	C	53	B
4	D	14	A	24	D	34	C	44	C	54	D
5	C	15	B	25	C	35	B	45	A	55	B
6	D	16	B	26	C	36	A	46	B	56	D
7	D	17	D	27	B	37	B	47	B	57	C
8	A	18	B	28	A	38	D	48	A	58	D
9	D	19	C	29	C	39	B	49	B	59	A
10	C	20	B	30	B	40	A	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Weather is the atmospheric condition of a place over a short stretch such as an afternoon, while climate is the average pattern built up over many years. The headline describes one afternoon's showers, so weather is the right word there; the textbook describes a steady long-term average, so climate is right there. The lure that swaps the two ignores the time-scale that separates them, and the idea that a single rainy afternoon equals climate confuses one event with a thirty-year average.

2. (C) Climate is the average atmospheric behaviour of a place taken over a long period, typically decades, so no single day's weather can alter it. Riya has mistaken a one-day event (weather) for the long-run pattern (climate). The claims that one rainy day or a heavier downpour could shift the climate keep the same time-scale error, and saying the two words are identical ignores that the only real difference between them is the length of time observed.

3. (D) Latitude controls how directly sunlight strikes a place and how much that directness changes through the year. Near the Equator the Sun stays nearly overhead all year, giving an even climate; farther north the angle swings widely between summer and winter, giving Delhi its big seasonal range. Altitude is not the driver here, the Equator is not meaningfully nearer the Sun (the difference is trivial against 150 million km), and Delhi certainly has an atmosphere.

4. (D) Earth is so far from the Sun that the tiny difference in distance between Equator and poles is negligible; what matters is the angle of the rays. At the Equator rays strike nearly straight down, concentrating heat on a small patch, while near the poles they slant and spread thinly over a wider area (and travel through more air). The "closer to the Sun" and "no air" answers are the classic misconceptions, and Earth's internal heat plays no role in surface climate here.

5. (C) As you climb, the air thins and holds heat less well, and you move away from the ground, which is the air's main heat source; so temperature drops with altitude. Town X, much higher, is therefore cooler despite the shared latitude. "Higher is nearer the Sun" repeats the equatorial-heat fallacy, sharing latitude does not erase the height difference, and altitude not latitude is what separates these two towns.

6. (D) Since both share a latitude, that factor is ruled out; the deciding factor is altitude. The peaks rise kilometres higher, where thin air and distance from the ground keep temperatures freezing year-round, while the low plains stay hot. The other options misattribute the difference to latitude, to the sea (the plains are not nearer the moderating sea than the peaks in any decisive way), or to imaginary equatorial winds.

7. (D) Water heats and cools far more slowly than land, so coastal places have their extremes softened, giving a small temperature range; this is continentality's mirror image. Chennai is coastal and Hyderabad is inland, so Chennai's range is smaller. Altitude is not the key (Chennai is low, not high), the two share a latitude so nearness to the Equator is not the cause, and Chennai obviously gets summer sunshine.

8. (A) Distance from the sea removes the ocean's smoothing effect, so land heats and cools sharply, widening the range; this is the heart of continentality. The first wrong option describes the opposite (sea + low land = small range), the equatorial option would give an even climate not a wide swing, and constant moist sea winds would moderate rather than widen the range.

9. (D) The monsoon is driven by the unequal heating of land and sea: land warms quickly in summer to form low pressure, drawing in moist sea air. The Moon's pull governs tides, not monsoon winds; rivers do not power the system; and a fixed equatorial temperature has nothing to do with the seasonal land-sea pressure flip that creates the monsoon.

10. (C) Air rushes from high pressure toward low pressure. The hot summer land forms a low-pressure zone, so the relatively cooler, high-pressure sea air flows inland and brings moisture; this is the south-west summer monsoon. High pressure over the land would push air out (the winter case), and the option pairing low pressure with outward dry winds reverses the direction air actually moves.

11. (A) Winds are named for the direction they come from; the summer rains are carried by winds arriving from the south-west, hence "south-west monsoon." It does not name a region the rain avoids, river flow direction, or the Sun's setting point. Naming by source direction is the same convention used for the north-east winter monsoon.

12. (D) The summer monsoon blows from sea to land, soaking up moisture over warm oceans and releasing it as rain; the winter monsoon blows from land to sea, so it is dry. The wrong options reverse the source of each wind, wrongly tie rain to coldness, or imagine rain dropping vertically from space rather than being carried in by moist sea air.

13. (A) The "burst" marks the arrival of the rain-bearing south-west winds, which switch a parched, scorching landscape to days of heavy downpour quite abruptly. The Sun does not stop shining, rivers do not evaporate upward to make the rain, and the burst is a summer (south-west) event, not the north-east winter monsoon arriving.

14. (A) The south-west monsoon enters at the southern tip in early June and progresses northward, blanketing the country by roughly mid-July. It does not march from north to south, it does not move from the land out to the seas, and it does not arrive everywhere simultaneously because it takes weeks to advance across India's length.

15. (B) The monsoon exists only because land heats and cools faster than the sea, creating the pressure differences that drive the winds. Erase that difference and there is no pressure gradient, so no monsoon. It cannot strengthen, shift to winter, or merely reverse, because all of those still require the very land-sea contrast that has been removed.

16. (B) Wind is air flowing down the pressure gradient, from high pressure to low pressure. The hot summer interior is low pressure, so the higher-pressure sea air flows in. Air does not move from low to high pressure, is not fixed to a north-south path, and does not simply chase the overhead Sun independent of pressure.

17. (D) In winter the land cools faster than the sea, becoming a high-pressure zone, while the relatively warmer sea is lower pressure; air therefore flows from land to sea, giving the dry north-east monsoon. The other options reverse the land-sea pressure roles or deny the pressure difference entirely, none of which would produce the observed offshore winter winds.

18. (B) Although offshore winds are mostly dry, those that travel over the Bay of Bengal absorb moisture and then shed it on the south-east coast as winter rain. A desert cannot supply moisture, the Himalayas lie to the north not in this wind's path to the south-east

coast, and the Arabian Sea is on the west, not the source for the north-east winter flow reaching Tamil Nadu.

19. (C) In summer the land is hotter (low pressure) and pulls sea air in; in winter the land is colder (high pressure) and pushes air out. This seasonal flip of the land-sea temperature and pressure pattern reverses the winds. Earth's tilt is fixed, mountains do not move, and the Moon's orbit is irrelevant to the monsoon.

20. (B) The retreating (autumn) season is when the south-west monsoon weakens and pulls back from the country, bridging the rainy season and the dry winter. It is not the burst (the arrival), it has nothing to do with rivers moving, and it follows rather than precedes the rains, so it is not the pre-monsoon heat build-up.

21. (A) The Indian year runs from the cool dry winter, into the hot dry pre-monsoon summer, into the rainy south-west monsoon when the burst arrives, and finally the retreating monsoon (autumn) as the rains withdraw before winter returns. The other orders break this cycle by placing the rains or the retreat out of their natural sequence.

22. (A) The monsoon so thoroughly governs Indian weather, farming and life that the rainy season stands out as its own distinct season, giving the familiar "fifth season." It was not borrowed from Europe (the reverse, India recognises the monsoon more finely), India does have a winter, and festivals follow the seasons rather than creating them.

23. (D) The six-ritu scheme reads the monsoon-dominated Indian year in finer detail, recognising transitions the four-season model lumps together. It is ancient rather than recent, it is built around the monsoon rather than ignoring it, and six categories cannot be a mere renaming of four.

24. (D) Seasons recur because Earth completes one revolution around the Sun each year while tilted on its axis, so each part of the year repeats. The Sun's output does not rise and fall seasonally, festival dates follow the seasons rather than commanding the winds, and river floods are a result of the seasons, not their cause.

25. (C) Winds carry the qualities of the land they cross; the loo passes over hot, dry deserts and arrives blistering and parched. Had it crossed the sea it would be moist, not dry; coming off the Himalayas it would be cold, not hot; and it is a surface wind, not heated by underground springs.

26. (C) Winds bring the character of wherever they have been; cold air off the snowy mountains arrives cold and chills the plains. Winds cannot change a place's latitude, raise its altitude, or relocate it nearer the sea; they simply transport the heat or cold and moisture they picked up en route.

27. (B) A microclimate is a small area whose conditions differ from the surroundings, like a shady, moist forest patch. A rain shadow is the dry leeward side of a mountain (a

different, larger phenomenon), a *ritu* is a season, and a monsoon is a seasonal wind system, none of which describes this small local pocket.

28. (A) Buildings and pavements store and release heat, creating a warm "urban heat island," a microclimate produced by human activity. It is not a rain shadow (which is a wind-and-relief effect on the dry side of mountains), not a season, and not a cyclone; it is simply a small, locally altered climate.

29. (C) Relief, the land's physical shape, both exposes the flat Thar to hot dry winds and lets the Himalayas block cold northern winds; this is topography at work. Latitude, sea distance and the Sun's position do not explain why a mountain wall shields one region while an open plain is exposed; the deciding factor here is the lay of the land.

30. (B) Ladakh-type areas sit in the rain shadow of the Himalayas, so they receive little rain (a desert), and their high altitude makes them bitterly cold; hence "cold desert." It is not a hot desert, its cold comes mainly from altitude rather than nearness to the pole, and far from being near the sea, its dryness comes from the mountain barrier cutting off moist winds.

31. (D) When moisture-laden monsoon winds are pushed up against the north-eastern hills, they cool, condense and unleash extraordinary rainfall, classic orographic rain. The Thar supplies dryness not rain, descending leeward air warms and dries (it does not rain), and these are warm moist summer winds, not cold Himalayan winter winds.

32. (A) The Western Ghats lift the incoming moist winds; the air cools and rains on the seaward (windward) slopes, so by the time it crosses to the Deccan it is dry, a rain shadow. The Deccan is not nearer the Equator, the coast is not higher than the Ghats, and rivers do not generate rainfall, so relief is the true cause.

33. (A) Rising air expands and cools; cool air holds less water vapour, so the excess condenses into cloud and rain on the windward slope. Rising air does not warm or hold more vapour, it does not dry without cooling, and air that has risen moves over the crest and sinks (warming and drying) on the far side, not back down the windward face.

34. (C) By the time air crosses the crest it has shed most of its moisture as windward rain, so the leeward side lies in a "shadow" of rain, dry. It has nothing to do with literal sunlight shadows or shadiness, and the leeward side is the one facing away from the moist wind, not into it.

35. (B) The heavy west-coast rain comes from moist winds being forced up the Ghats, cooling and condensing; remove the barrier and the winds would sweep inland without rising sharply, so the coast would get less rain. It would not get more rain (the lifting is what wrings out the water), would not freeze, and certainly would not stay unchanged.

36. (A) Srinagar is both far north (high latitude, weaker, more slanting sunlight) and high in the mountains (low temperatures with altitude), and these stack up to make it cold. It is not at a lower latitude, it is nowhere near the Equator, and it is not bathed in moderating sea winds, so the correct pair is latitude plus altitude.

37. (B) Mumbai is coastal and Nagpur is inland, so the slow-changing sea evens out Mumbai's temperatures, giving milder extremes; both statements are true and the reason is exactly why. The other choices wrongly sever the link or deny one statement, but the sea's slow heating is precisely the mechanism behind Mumbai's gentler climate.

38. (D) Near the Equator the Sun's overhead position barely changes through the year, so heating stays steady and the seasonal temperature change is small; the reason genuinely explains the assertion. The distractors either break the true causal link or wrongly deny one of the statements, but here both are true and connected.

39. (B) It is true that the winter monsoon is mostly dry, but the stated reason is backwards: winter winds blow from the cold high-pressure land toward the sea, which is exactly why they are dry. So the assertion is true while the reason is false, and the options that accept the reason or deny the dry-winter fact are wrong.

40. (A) With latitude held equal, the remaining controlling factors are altitude and exposure to the sea's moderating influence, so those are the sensible things to check. Distance to the Sun and the Moon's path do not vary meaningfully between two nearby cities, and festivals have no effect on temperature range.

41. (C) The pattern, drenched windward coasts and hills with drier leeward interiors, is explained by moist monsoon winds being forced up over relief and raining there, then arriving drier beyond. Rainfall does not simply track latitude, height alone does not guarantee rain (it depends on windward exposure), and inland is generally drier, not wetter, so relief plus wind direction is the unifying idea.

42. (B) Most Indian farming depends on monsoon rain, so a failed monsoon dries the soil, shrinks harvests and pushes food prices up, sometimes forcing migration. A failed (weak) monsoon means less water, so floods and bumper harvests are the opposite of what happens, and it neither warms winters nor erases the seasons.

43. (C) When the periodic Pacific warming (El Niño) shifts winds and weakens the monsoon, India tends to get less rain, hurting crops and water supplies. A weakened monsoon brings shortfall, not record floods; it does not alter mountain heights; and it has nothing to do with whether the sea moderates the coast.

44. (C) Indian agriculture relies heavily on the monsoon, so when the rains fail, crops fail, food becomes scarce and dear, and rural incomes fall, all of which strain the economy. Festivals are not legally cancelled, the sea's moderating role does not switch off, and the Himalayas do not block trade because of a dry year.

45. (A) El Niño is the periodic warming of Pacific waters that can shift winds and weaken the Indian monsoon. The loo is a hot summer land wind, a rain shadow is the dry leeward side of mountains, and the burst is the sudden onset of monsoon rain, none of which is the Pacific warming described.

46. (B) A sea breeze forms when sun-warmed land creates low pressure and pulls in cooler sea air, exactly the land-sea heating contrast that powers the monsoon, just on a country-sized scale. A rolling ball, the Moon's tides and a flowing river involve different forces and have nothing to do with land-sea pressure differences.

47. (B) These harvest and seasonal festivals fall when crops are sown or gathered, showing how the monsoon-driven climate steers farming and hence cultural life. Festivals do not command the rains, climate clearly does shape culture (that is the very point), and the festivals follow the farming calendar rather than mere latitude.

48. (A) Mausim means "season," and "monsoon" properly names the seasonally reversing winds; the rain is a consequence of those winds, not the word's meaning. It does not name the rain by itself, nor the ocean, nor the six r̥itus, so the seasonally shifting winds are the correct sense.

49. (B) A cyclone is built around an intense low-pressure centre into which surrounding air and moisture spiral, feeding the storm. High pressure would push air out (the opposite of a cyclone), equal pressure produces no winds at all, and cyclones form over warm seas, not from winter cold over land.

50. (D) The calm, clear "eye" at a cyclone's centre is deceptive; once it passes, the violent winds of the far eyewall strike from the opposite side, so people must stay sheltered. The lull does not mean the storm is over or harmless, and although the calm eye feels safe, the dangerous part is the surrounding eyewall, not the eye itself.

51. (C) The fiercest winds whirl in the eyewall, the thick cloud band ringing the calm eye. The eye itself is deceptively still, the violent winds are within the storm rather than far outside it, and they certainly do not appear after the cyclone has gone.

52. (A) Greenhouse gases trap heat; adding more from burning fuels thickens the blanket and traps even more, warming the Earth. They do not let heat escape (that would cool it), they do not mainly block incoming sunlight, and they do not remove the atmosphere, so added greenhouse gas means more trapped heat and warming.

53. (B) Burning coal releases carbon dioxide, which traps extra heat, warming and shifting the climate, which can disrupt the monsoon and cause crops to fail; that is the correct chain. The alternatives go wrong by cooling the sea, freezing rivers or growing forests, none of which follows from burning coal, which adds greenhouse gas and warms.

54. (D) Mitigation tackles the cause by reducing greenhouse-gas emissions and slowing warming, while resilience prepares communities to cope with impacts like floods or droughts. The first wrong option swaps the two definitions, the second over-narrows both to tree-planting, and the third mischaracterises each as ignoring half the problem.

55. (B) Mitigation reduces the greenhouse gases that drive warming, so switching to renewable energy and planting trees (which absorb carbon dioxide) qualify. Clearing forests, burning more coal and wasteful consumption all add greenhouse gases, making the problem worse rather than mitigating it.

56. (D) The rapid warming since the industrial era is driven mainly by human activity, above all burning fossil fuels, which adds greenhouse gases. Natural processes and occasional volcanoes have not caused this fast rise, and the Earth is not drifting away from the Sun, so human emissions are the principal cause.

57. (C) Without the natural greenhouse effect, Earth would be far too cold; greenhouse gases trap part of the Sun's heat and keep the planet comfortably warm. They do not block all sunlight or freeze the planet (the opposite of warming), and they do not directly create the monsoon, which arises from land-sea heating.

58. (D) Air forced up the windward slope cools and rains there, then descends dry on the leeward side, so a short crossing can take you from wet to dry, the rain shadow. A few kilometres cannot change latitude or nearness to the Equator meaningfully, and soil type does not control rainfall, so relief and wind direction are the cause.

59. (A) Because India stretches across many degrees of latitude, from sea level to the high Himalayas, and from coast to deep interior, its controlling factors vary widely, producing many climates at once. Calendars do not cause climate, the Sun does not single out states, and weather is not controlled by political boundaries; geographic variety is the real reason.

60. (C) Ladakh is a cold desert because of high altitude plus the Himalayan rain shadow; Mumbai's mild range comes from the moderating sea; Cherrapunji's deluge comes from relief forcing moist winds to rise; and Kanniyakumari stays warm because of its low (near-equatorial) latitude. The other matchings misassign these, for instance crediting Ladakh's cold to low latitude or Mumbai's mildness to altitude, which contradicts each region's true controlling factor.